Lecture 1
Probability and Bayesian Statistics

From Bayesian statistics to probabilistic machine learning

Probabilistic Machine Learning 2027-1

2026-10-05

Section 1

A familiar modeling language

The Beta-Binomial model

Observed

the coin tosses $y_1, ..., y_N$

Unknown

the coin bias θ

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta), \quad i = 1, ..., N$$

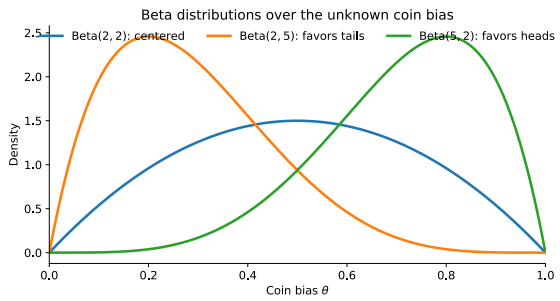
Code heads as $y_i = 1$ and tails as $y_i = 0$

$$N_1 = \sum_{i=1}^{N} y_i, \quad N_0 = N - N_1$$

- A statistical model has data and unknown parameters
- How could observed data (y) have been generated.
- What does it tell us about the parameters (θ)
- N_1 and N_0 count heads and tails in the observed sample

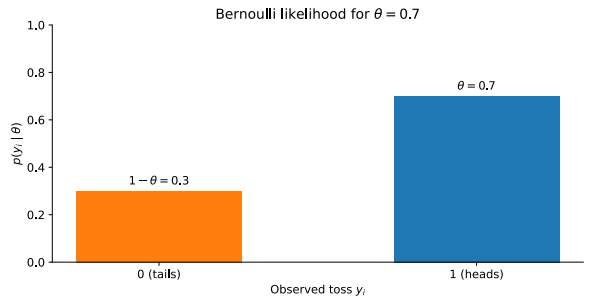
Models are built out of distributions

$$\theta \sim \text{Beta}(\alpha, \beta)$$



figures/coin_model__beta_distribution.svg

$$y_i \sim \text{Bernoulli}(\theta)$$
$$i = 1, \ldots, N$$



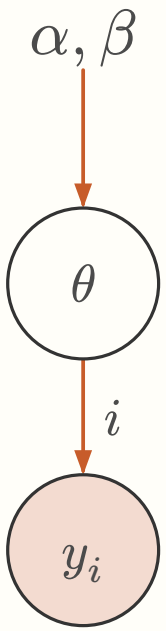
figures/coin_model__bernoulli_likelihood.svg

- Beta for continuous variables on $[0, 1]$.
- Bernoulli for each yes/no outcome; Binomial for the count N_1.
- Distributions are parameterized families

Models can be drawn as graphs

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$



- Nodes represent quantities.
  - Shading marks observed data.
- Arrows encode conditional dependence.
  - We will not use Kruschke diagrams

In this course, graphs will allow us to factor complex problems into smaller, more tractable problems

Bayes' rule combines prior and likelihood

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes' rule

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

$$\propto p(y \mid \theta)p(\theta)$$

- **Prior:** $p(\theta)$
- **Likelihood:** $p(y \mid \theta)$
- **Posterior:** $p(\theta \mid y)$
- We explored the posterior of θ
  - Treating $p(y)$ as a normalizing constant

Conjugacy permits simple calculations

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes rule

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

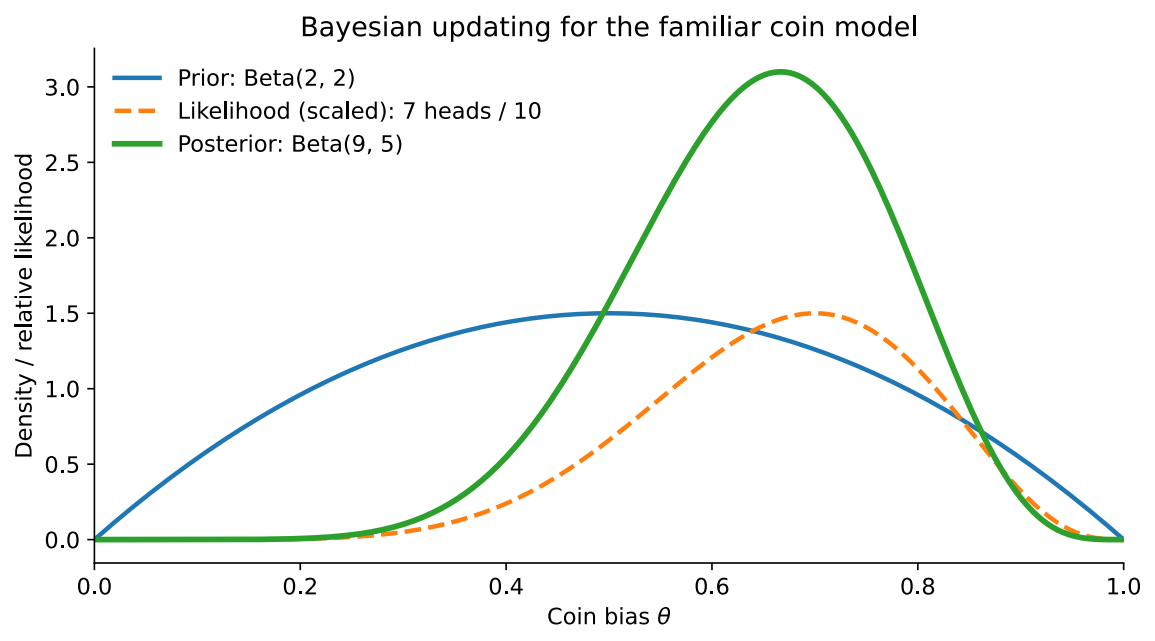
Applying Bayes' rule to our model

Prior	$p(\theta) = \text{Beta}(\alpha, \beta)$
Likelihood	$p(y \mid \theta) \propto \theta^{N_1}(1-\theta)^{N_0}$
Posterior	$p(\theta \mid y) = \text{Beta}(\alpha + N_1, \beta + N_0)$

Conjugacy

The prior and likelihood are conjugate when the posterior belongs to the same distribution family as the prior.

The posterior is a distribution representing uncertainty



Bayesian updating for the familiar coin model

`figures/coin_model__prior_likelihood_posterior.svg`

- The prior and the posterior are both probability distributions
- Their width expresses the uncertainty
- The uncertainty in the posterior is less than in the prior

Summary: A familiar modeling language

- Probabilistic models link observed data to parameters
- Graphs show model variables and dependencies
- Bayes' rule combines prior and likelihood into a posterior
- Conjugacy makes some updates analytic
- Posterior width represents remaining uncertainty
- Posterior uncertainty narrows with more data

Section 2

From statistics to machine-learning problems

Same probabilistic language, different organizing questions

Statistics focuses on the unknown parameters

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

- Get data → build a model → apply Bayes' rule → interpret θ
  - For many scientific questions, that posterior **is** the answer
- Statistics arose from an interest in the underlying parameters in nature

Machine learning grew out of computer science



- A computational task is an **input/output contract**
  - Produce a function f that maps input x to output y
- Implementation is secondary to the requirement
- Multiple implementations may be possible
- Machine learning developed in this context

Early machine learning grew out of classification problems

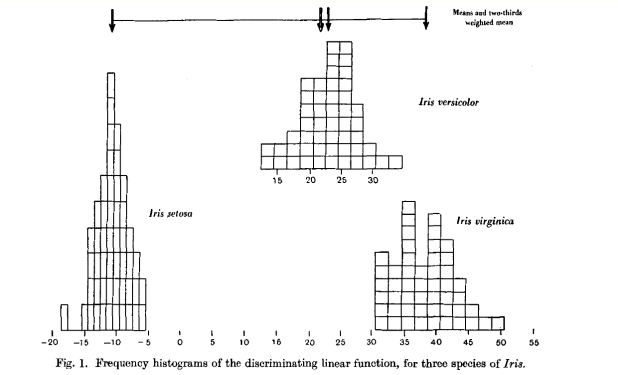


Fig. 1. Frequency histograms of the discriminating linear function, for three species of *Iris*.

Fisher (1936), Fig. 1, The Use of Multiple Measurements in Taxonomic Problems

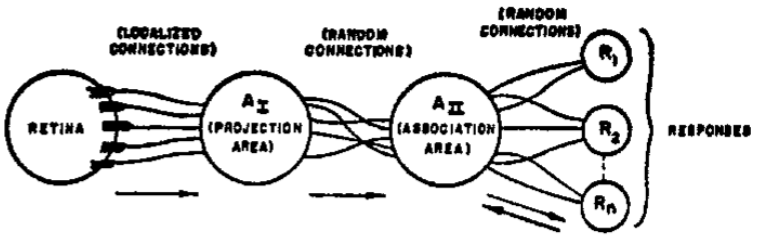
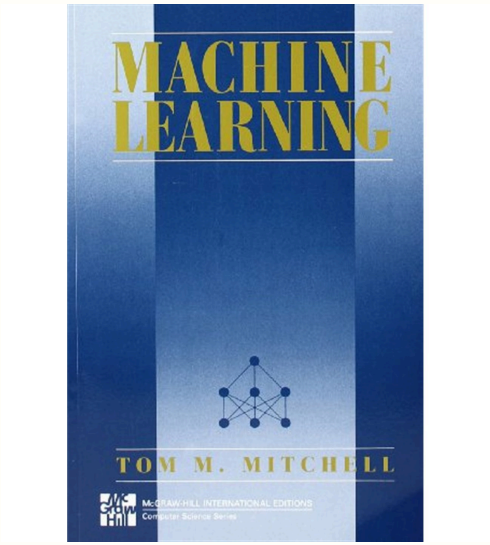


FIG. 1. Organization of a perceptron.

Rosenblatt (1958), Fig. 1, The Perceptron

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input $\rightarrow$ class or response
- Machine learning asks how data can produce the required function f

Machine learning defined



Tom M. Mitchell, Machine Learning (1997), cover

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. ,

Training	**Task**	**Evaluation**
$f_T : \mathrm{E} \rightarrow f_A$	$f_A : \mathrm{In} \rightarrow \mathrm{Out}$	$f_P : (\mathrm{In}, \mathrm{Out}) \rightarrow \mathrm{P}$
E: Experience	f_A: Task function	P: Performance

- Machine learning is a mapping from $(E, f_P) \rightarrow f_T \rightarrow f_A$
  - That maximizes f_A performance according to P

Supervised experience supports prediction

Experience: paired examples

$$f_T : \mathrm{E} \quad \to \quad f_A$$

$$f_T : D = \{(x_i, y_i)\}_{i=1}^{N} \quad \to \quad f_D$$

paired examples → learned predictor

Task: prediction

$$f_A : \mathrm{In} \quad \to \quad \mathrm{Out}$$

$$f_D : x_* \quad \to \quad \hat{y}_*$$

new input → predicted output

- Experience is a training set $D = \{(x_n, y_n)\}$
  - x: called feature, covariate, or predictor
  - y: called label, target, or response
- Task is to produce a predictor f_D that maps new x_* to correct y_*
- Performance is measured by a loss, introduced shortly

Classification: Iris flowers


Setosa


Versicolor


Virginica

Table 1.1: subset of the Iris design matrix

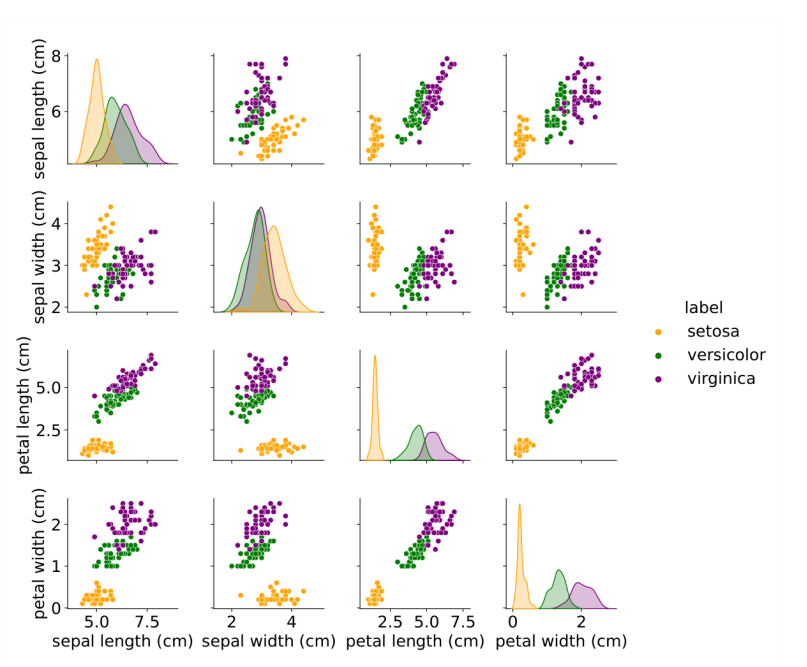
index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
dots	dots	dots	dots	dots	dots
50	7.0	3.2	4.7	1.4	Versicolor
dots	dots	dots	dots	dots	dots
149	5.9	3.0	5.1	1.8	Virginica

sl: sepal length; sw: sepal width; pl: petal length; pw: petal width

Murphy (2025), Book 1, Fig. 1.1 and Table 1.1

- Classification: $y \in \{1, ..., C\}$
  - The classes are unordered and mutually exclusive
- For the Iris dataset
  - $y \in \{\text{Setosa}, \text{Versicolor}, \text{Virginica}\}$
  - $x \in \mathbb{R}^4$

Exploratory data analysis



Murphy (2025), Book 1, Fig. 1.3; generated by iris_plot.ipynb

- Training set
  - 150 labeled examples
  - 50 in each class
- Pair plots shows every pair of features
  - Diagonal shows marginal distributions
- Always inspect data before modeling
  - Structure
  - Outliers
  - Missing values

Performance: empirical risk minimization

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} L(x_n, y_n; \theta)$$

$$\hat{\theta} = \arg\min_{\theta} L(\theta)$$

Given samples $D = \{x^1, ..., x^N\}$ from $p(X)$, where $X \in \mathbb{R}$, the empirical distribution is $\hat{p}_N(x) = \frac{1}{N} \sum_{n=1}^{N} \delta_{x^n}(x)$.

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y
- The empirical risk: loss evaluated on the training set
- Empirical risk minimization: minimizing the empirical loss

A loss function

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$\ell(y, f(x; \theta)) = \ell_{01}(y, \hat{y}) = \mathbb{I}(y \neq \hat{y})$$

$$\mathbb{I}(e) = \begin{cases} 1 \text{ if } e \text{ is true} \\ 0 \text{ if } e \text{ is false} \end{cases}$$

- A loss function assigns a cost to a prediction when the true label is known
- The zero-one loss is one for a misclassification and zero for a correct classification
- $\mathbb{I}(e)$ is an indicator function: it evaluates whether condition e holds

A probabilistic perspective

$$p(x \mid y = c, \theta)$$

$$p(y = c \mid x, \theta) \propto p(x \mid y = c, \theta) p(y = c)$$

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} p(y = c \mid x; \theta)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions
- Bayes' rule converts those ingredients to posterior class probabilities
- A possible classifier: choose the class with maximum a posteriori probability

Task, experience, and performance

Task

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

Experience

What data, feedback, or interaction is available?

$$D = \{(x_n, y_n)\}_{n=1}^{N}$$

150 labeled flowers

Performance

How is success evaluated?

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}(y_n \neq f(x_n; \theta))$$

zero-one loss / misclassification rate

- **Task:** predict the Iris species from four sepal and petal measurements
- **Experience:** 150 labeled flowers, with 50 examples from each species
- **Performance:** mean zero-one loss, the training misclassification rate

Summary: From statistics to machine learning

- Statistics and computer science
  - Statistics focused on the values of unknown parameters
  - Computer science develops functions that map inputs to outputs
- Machine learning is a branch of computer science
  - Where the function is learned from experience
- A learning problem is defined by three ingredients: task, experience, and performance
  - One canonical pairing uses supervised experience $D = \{(x_n, y_n)\}$ for prediction
  - The experience regime and the task are related, but they are not the same taxonomy

Section 3

Classes of machine learning problems

Experience regimes, task classes, and performance criteria

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

Unsupervised

$$D = \{x_n\}$$

Self-supervised

$$D = \{x_n, f(x_n)\} \text{ or } \{f(x_n), x_n\}$$

Reinforcement / interaction

$$\begin{aligned} \text{decision} &: x(t) \to a(t) \\ \text{action} &: (x(t), a(t)) \to x(t+1) \\ \text{reward} &: x(t+1) \to r(t+1) \end{aligned}$$

> Experience describes what the learner gets to observe

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

Prediction	Map inputs to outputs
Generation	Produce new examples
Discovery	Find a representation of the data
Action	Choose actions to achieve desired outcomes

There are many combinations of experience and task

Experience	Prediction	Generation	Discovery	Action
Supervised	Classification	—	—	Imitation learning
Unsupervised	—	Density modeling	Clustering / PCA	—
Self-supervised	Next-token prediction	Language generation	Representation learning	—
Reinforcement / interaction	—	—	—	Policy learning

- Not every type of experience supports every task.
- Common pairings do exist.

Summary: Two taxonomies, three ingredients

- **Learning problem** = task + experience + performance
- **Experience regimes:** supervised, unsupervised, self-supervised, interactive
- **Task classes:** prediction, generation, discovery, action
- Probabilistic modeling provides a unifying language

Section 4

From datasets to learning problems

The same dataset can pose different learning problems

Popular datasets

Iris





AGE
TASKS

Since 1936 – about 90 years
species prediction; clustering; visualization

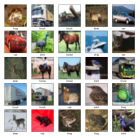
MNIST



AGE
TASKS

Since 1998 – about 28 years
digit classification; representation learning; generation

CIFAR-10



AGE
TASKS

Since 2009 – about 17 years
object classification; robustness; generation

IMDB movie reviews

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

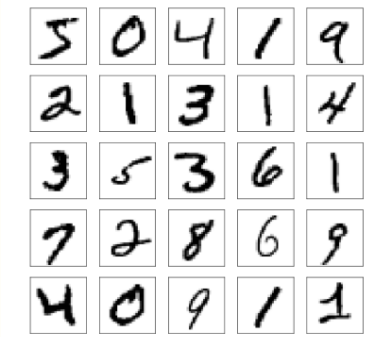
AGE
TASKS

Since 2011 – about 15 years
sentiment classification; text representation

Murphy (2025), Book 1, Figs. 1.1, 1.12a, 1.13b, and Table 1.3

- Machine learning is built around a small number of canonical datasets
  - These four datasets are commonly used
- Each dataset supports multiple tasks
- Datasets support benchmarks that make comparisons easy

MNIST: the same taxonomy, one dataset



$$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$$

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

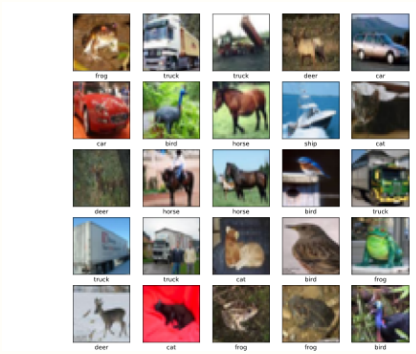
$$\hat{z} = f_D(x_*)$$

Generation

$$x_* \sim p_D(x \mid c)$$

Murphy (2025), Book 1, Fig. 1.12a

CIFAR-10: higher-dimensional inputs, the same questions



Prediction

$$\hat{y} = f_D(x_*)$$

$$L(\theta; D_{\text{train}}) \quad (D_{\text{train}}) \rightarrow L(\theta; D_{\text{test}}) \quad (D_{\text{test}})$$

$$x_n \in \mathbb{R}^{32\times32\times3}, \quad y_n \in \{1, \ldots, 10\}$$

Murphy (2025), Book 1, Fig. 1.13b

IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

$$\hat{z} = f_D(x_*)$$

Generation

$$w_t \sim p_D(w_t \mid w_{<t})$$

Murphy (2025), Book 1, Table 1.3

Learning problems become probabilistic queries

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

- Every task invokes probability models: $p_D(y|x)$, $p_D(x|c)$, $p_D(z|x)$
- Performances are expectations over future cases: $f_P = E[\cdot]$
- Probability models provide a common language for all four task schemas

Summary: From datasets to learning problems

- A dataset is not a task
- A learning problem names variables and a query
  - prediction: summarize $p_D(y_* \mid x_*)$
  - discovery: infer $p_D(z_* \mid x_*)$
  - generation: sample from $p_D(x \mid c)$

Section 5

Probability for machine learning

From learning problems to exact probabilistic operations

Large learning problems still ask familiar probability questions

Observed
x_{obs}: features, measurements, context

Latent
z: state, representation, missing value

Target
y_*: prediction, sample, decision

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

- The variables may be high-dimensional, latent, or only partly observed
- The task determines which quantity we need from the joint distribution
- The probability language is familiar even when the model is large

A few key operations underlie many machine learning tasks

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

Summarize

$s(x_{\text{obs}}) = \iint g(y_*) p_D(z, y_* \mid x_{\text{obs}}) \, \mathrm{d}z \, \mathrm{d}y_*$

weighted summary of the target

Marginalize

$p_D(y_* \mid x_{\text{obs}}) = \int p_D(z, y_* \mid x_{\text{obs}}) \, \mathrm{d}z$

average out the latent variable

Condition

$p_D(z \mid x_{\text{obs}}) = \frac{p_D(x_{\text{obs}}, z)}{\int p_D(x_{\text{obs}}, z') \, \mathrm{d}z'}$

infer the latent variable from observations

- Almost all machine learning tasks require an integral
- Latent variables are inferred or averaged out
- Much methodology addresses computing these integrals efficiently

Generalize the operations using generic random variables

X and Y have joint density $p(x, y)$

Summarize

$$E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$$

Marginalize

$$p(x) = \int p(x, y)\,\mathrm{d}y$$

$$p(y) = \int p(x, y)\,\mathrm{d}x$$

Condition

$$p(x \mid y) = \frac{p(x, y)}{p(y)}$$

- The same operations apply to any pair of random variables X and Y
- Lowercase x and y are possible values; conditioning fixes $Y = y$
- The normalizer is another marginal

Section 6

Joint Gaussian distributions and geometry

Describe and transform joint uncertainty with a mean and covariance

Gaussians: a concrete example

One variable

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

D coordinates

$$\mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T\Sigma^{-1}(\boldsymbol{x}-\mu)\right)}{(2\pi)^{\frac{D}{2}}\ |\Sigma|^{\frac{1}{2}}}$$

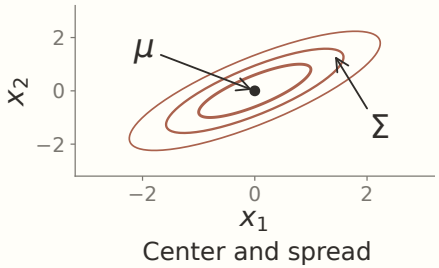
Murphy, Book 2 §2.3.1.1, Eq. 2.71; scalar case: $D = 1$

- One variable: mean μ and variance σ^2
- Mean vector $\mu \in \mathbb{R}^D$ locates the cloud; covariance $\Sigma \in \mathbb{R}^{D\times D}$ sets its shape
- Gaussian means, marginals, and conditionals have closed forms

The three operations in multivariate Gaussian systems

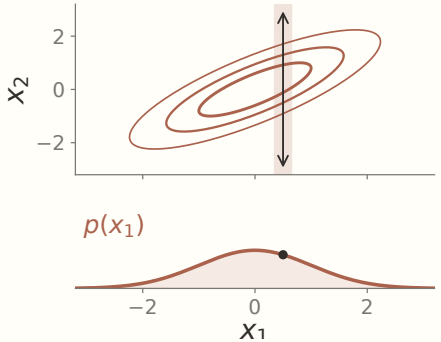
One joint Gaussian: $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma)$

Summarize



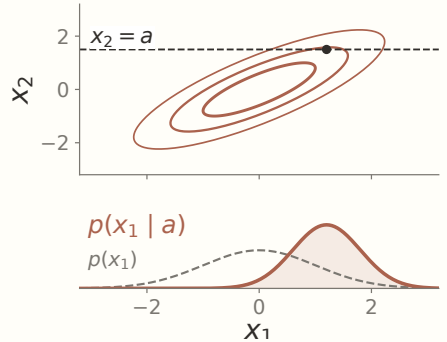
$$\mu = \int \boldsymbol{x} p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}$$

Marginalize



$$p(x_1) = \int p(x_1, x_2) \, \mathrm{d}x_2$$

Condition



$$p(x_1 \mid x_2 = a) = \frac{p(x_1, a)}{\int p(u, a) \, \mathrm{d}u}$$

Figures: `gaussian_geometry_figures.ipynb`

- **Summarize:** describe the center and spread with μ and Σ
- **Marginalize:** integrate over the coordinate we ignore
- **Condition:** fix an observed coordinate and normalize the slice

Covariance summarizes joint variation

For a vector $\boldsymbol{x}$ with mean $\mu = E[\boldsymbol{x}]$

$$\Sigma_{ij} = \operatorname{Cov}[X_i, X_j] = E\big[(X_i - \mu_i)(X_j - \mu_j)\big]$$

$$\Sigma = E\big[(\boldsymbol{x} - \mu)(\boldsymbol{x} - \mu)^T\big] = E[\boldsymbol{x}\boldsymbol{x}^T] - \mu\mu^T$$

For three coordinates

$$\Sigma = \begin{bmatrix} V[X_1] & \operatorname{Cov}[X_1, X_2] & \operatorname{Cov}[X_1, X_3] \\ \operatorname{Cov}[X_2, X_1] & V[X_2] & \operatorname{Cov}[X_2, X_3] \\ \operatorname{Cov}[X_3, X_1] & \operatorname{Cov}[X_3, X_2] & V[X_3] \end{bmatrix}$$

Murphy, Book 1 §3.1.1, Eqs. 3.1–3.4

- Positive / negative covariance: deviations tend to have the same / opposite signs
- Diagonal entries are variances; off-diagonal entries are covariances
- Zero covariance does not imply independence

Correlation separates direction from units

$$\rho_{ij} = \frac{\mathrm{Cov}[X_i, X_j]}{\sigma_i \sigma_j}, \quad -1 \leq \rho_{ij} \leq 1$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 \end{bmatrix}$$

Murphy, Book 1 §3.1.2, Eq. 3.7; Book 2 §2.3.1.1, Eq. 2.73

- ρ_{ij} describes the direction and strength of linear association
- $\sigma_1, \sigma_2, \sigma_3$ set the coordinate scales

Affine maps carry Gaussian means and covariances

$$\boldsymbol{y} = A\boldsymbol{x} + b$$

$$E[\boldsymbol{y}] = A\mu + b$$

$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

Covariance proof

Center: $\boldsymbol{d} = \boldsymbol{x} - \mu$

$$\begin{aligned}
\boldsymbol{y} &= A(\boldsymbol{d} + \mu) + b \\
&= A\boldsymbol{d} + (A\mu + b) \\
\boldsymbol{y} - E[\boldsymbol{y}] &= A\boldsymbol{d} \\
\mathrm{Cov}[\boldsymbol{y}] &= E[(A\boldsymbol{d})(A\boldsymbol{d})^T] \\
&= AE[\boldsymbol{d}\boldsymbol{d}^T]A^T \\
&= A\Sigma A^T
\end{aligned}$$

Murphy, Book 1 §3.1.1, Eq. 3.5

- A shift changes the mean but not the covariance
- A linear map changes the scale, orientation, and dimension of variation

Mahalanobis distance uses the cloud's own scale

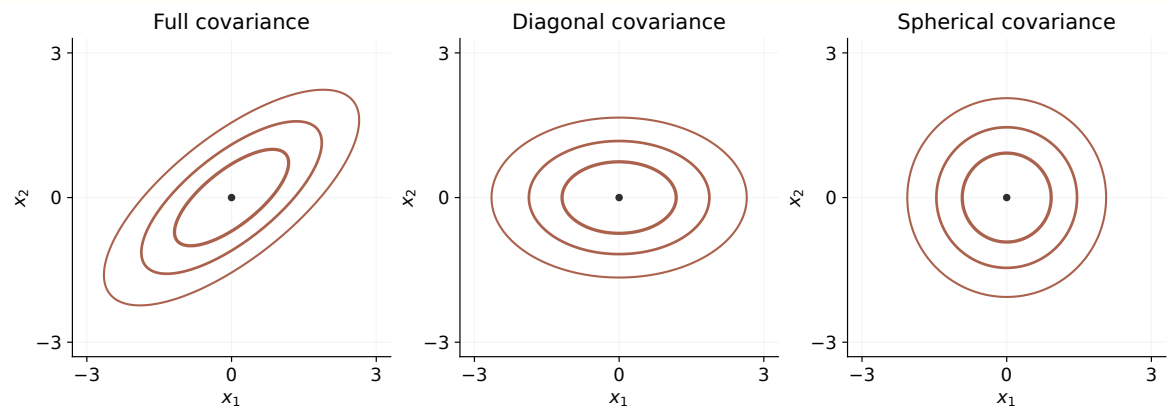
$$d_\Sigma(\boldsymbol{x}, \mu)^2 = (\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$\Sigma = \sigma^2 I_D \quad \rightarrow \quad d_\Sigma^2 = \frac{\|\boldsymbol{x} - \mu\|^2}{\sigma^2}$$

Murphy, Book 2 §2.3.1.1, Eq. 2.72

- Equal Mahalanobis distances trace equal-density contours
- With spherical covariance, it becomes squared Euclidean distance in standard-deviation units

Covariance controls the contour shape

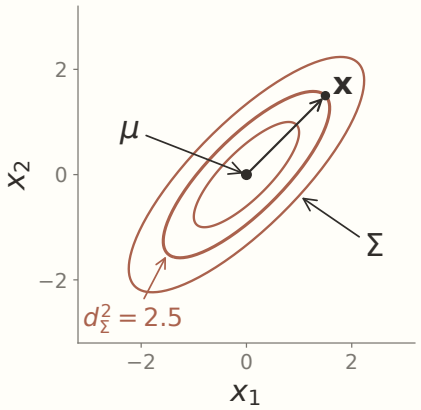


Course rendering of Murphy, Book 2, Fig. 2.8

- A full covariance can rotate the principal axes
- Diagonal covariance removes linear correlation in these coordinates
- Spherical covariance assigns the same variance in every direction

Summary: Joint Gaussian geometry

$$\boldsymbol{x} \sim \mathcal{N}(\mu, \Sigma)$$



Location: μ is the center

Variation: Σ sets scale and orientation

$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}$ removes units

Distance: $d_\Sigma^2 = (\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$

One contour: equal distance and equal density

Affine maps: $\boldsymbol{y} = A\boldsymbol{x} + b \quad \rightarrow \quad \mu_y = A\mu + b, \quad \Sigma_y = A\Sigma A^T$

Figure: `gaussian_geometry_figures.ipynb`

- Mean and covariance determine a Gaussian's joint law and geometry
- Next: ignore some coordinates, or condition on observed ones

Section 7

Marginals and conditionals of a Gaussian

Ignore coordinates or use observations to update uncertainty

Partition the variables and their parameters together

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad \Sigma_{21} = \Sigma_{12}^T$$

Murphy, Book 2 §2.3.1.3, Eq. 2.75

- x_1 and x_2 can each be vectors
- The diagonal blocks describe within-group variation
- The cross blocks describe variation between groups

Marginalization keeps the matching blocks

$$\boldsymbol{x} \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = P\boldsymbol{x}$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

$$P\mu = \mu_1$$

$$P\Sigma P^T = [I \;\; 0][\Sigma_{11} \;\; \Sigma_{12}] \begin{bmatrix} I \\ 0 \end{bmatrix}$$

$$= \Sigma_{11}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Start at the prior mean μ_1
- Respond to the innovation $x_2 - \mu_2$
- Cross-covariance and observed variance determine the response

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

Given x_2, the mean $\mu_{1|2}$ is constant

$$\Sigma_{1|2} = \mathrm{Cov}[x_1 \mid x_2] = \mathrm{Cov}[r]$$

$$\Sigma_{11} = B\Sigma_{22}B^T + \mathrm{Cov}[r]$$

$$\Sigma_{1|2} = \Sigma_{11} - B\Sigma_{22}B^T$$

$$= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values

The bivariate formula gives intuition

$$x_1, x_2 \in \mathbb{R}$$

$$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \sigma^2_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\sigma_{1|2} = \sigma_1 \sqrt{1 - \rho^2}$$

Murphy, Book 1 §3.2.4; Book 2 §2.3.1.3

- Positive ρ moves the estimate in the observed direction
- $\rho = 0$ leaves both mean and variance unchanged
- Larger $|\rho|$ leaves less conditional variance

One observed coordinate can change both the estimate and its uncertainty

$$x_1|2 \sim \mathcal{N}\Big(\mu_1 + \rho\tfrac{\sigma_1}{\sigma_2}(x_2 - \mu_2), (1-\rho^2)\sigma_1^2\Big)$$

$$x_1 \sim \mathcal{N}(0,1), \quad x_2 \sim \mathcal{N}(0,1), \quad \rho = 0.8$$

$$x_2 = 1.5$$

$$x_1 \mid x_2 = 1.5 \sim \mathcal{N}(0.8(1.5), 1 - 0.8^2) = \mathcal{N}(1.2, 0.36)$$

Worked example of Murphy, Book 2 §2.3.1.3, Eq. 2.78

- The observed value lies above its mean, so the conditional mean rises
- The variance falls from 1 to 0.36

The Schur complement is the residual covariance

Conditional Gaussian

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$S = \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$(x_1 \mid x_2) \sim \mathcal{N}\big(\mu_{1|2}, S\big)$$

Centered residual

$$r = x_1 - \mu_{1|2}$$
$$r \sim \mathcal{N}(0, S)$$

The residual r is independent of x_2

Remove the part predicted by x_2:

$$\mathrm{Cov}\left[\begin{pmatrix} r \\ x_2 \end{pmatrix}\right] = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.91, 2.102–2.103

- S measures the uncertainty left after subtracting the conditional mean
- Block elimination leaves S: the **Schur complement of Σ_{22} in Σ**

The joint quadratic separates into conditional and marginal parts

$$p(x_1, x_2) = p(x_1 \mid x_2)p(x_2)$$

$$(\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$= r^T S^{-1} r + (x_2 - \mu_2)^T \Sigma_{22}^{-1} (x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.94–2.103

- The first quadratic describes x_1 after seeing x_2
- The second describes the marginal distribution of x_2

Summary: Marginals and conditionals of a Gaussian

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Ignore x_2: marginalize

$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

Observe $x_2 = a$: condition

$$(x_1 \mid x_2 = a) \sim \mathcal{N}\left(\mu_{1|2}, S\right)$$
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2)$$
$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1: Gaussian marginals and conditionals

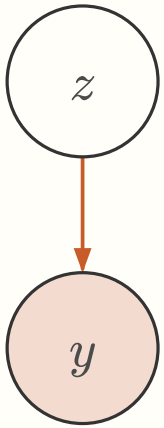
- Conditioning gives a mean and remaining uncertainty: $S = \Sigma_{1|2}$
- The observed value changes the mean; S depends only on the covariance blocks
- If $\Sigma_{12} = 0$, observing x_2 gives the same law as ignoring it

Section 8

Linear Gaussian systems

Inference from noisy linear measurements

A linear Gaussian system is a model of state and measurement



Latent state

Observed

A prior and a measurement law

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$

$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

$$p(z, y) = p(z)p(y \mid z)$$

$$y = Wz + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega)$$

Noise ε is independent of the state z.

Murphy, Book 2 §2.3.2, Eqs. 2.118–2.119

- This model specifies how an unknown state generates a noisy measurement
- Here the parameters $\mu_z, \Sigma_z, W, b, \Omega$ are given

One linear Gaussian system supports different questions

$$z \sim \mathcal{N}(\mu_z, \Sigma_z), \quad y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

Condition

$$p(z \mid y) = \frac{p(y \mid z)p(z)}{p(y)}$$

Summarize

$$E[g(z) \mid y] = \int g(z)p(z \mid y)\,\mathrm{d}z$$

Marginalize

$$p(y) = \int p(y \mid z)p(z)\,\mathrm{d}z$$

Murphy, Book 2 §2.3.2

- Conditioning asks what we can learn from an observed measurement
  - What is the distribution of the hidden state after seeing a measurement?
- Summarizing turns a distribution into a quantity
  - What is our estimate of the hidden state?
- Marginalization averages over an unknown variable to predict another
  - What is the distribution of a measurement before it is observed?

Inference often combines the operations

Noisy measurement		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y)$	$\rightarrow$	$\hat{z} = E[z \mid y]$

Murphy, Book 2 §2.3.2.2–2.3.2.3, Fig. 2.10

- Conditioning gives the posterior distribution over plausible states
- Summarize the posterior by its mean
- The operations combine: first obtain a distribution, then extract a quantity

A linear Gaussian model is also a joint Gaussian

Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y = Wz + b + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, \Omega)$$

Noise ε is independent of z

Joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_z \\ W\mu_z + b \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_z & \Sigma_z W^T \\ W\Sigma_z & W\Sigma_z W^T + \Omega \end{pmatrix}$$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- The affine-Gaussian rules construct $p(z, y)$ from the model
- Stack the state mean and the predicted measurement mean
- Independent noise adds Ω to the measurement covariance

Bayes' rule for Gaussians

Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z$, $x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

Bayes' rule as an update formula

Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right) \quad \mu_y = W\mu_z + b$$

$$\Sigma_{zz} = \Sigma_z \quad \Sigma_{zy} = \Sigma_z W^T$$

$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Posterior after observing y

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}(y - \mu_y)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Update the posterior mean

$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$

$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$

- K is the gain: it sets the size of the correction
- Read Bayes' rule as an update:
  - Prior estimate μ_z + gain K $\times$ prediction error $y - \mu_y$.

The update is weighted by relative uncertainty

From Bayes' rule

$\mu_{z|y} = \mu_z + K(y - \mu_y)$
$K = \frac{\Sigma_{zy}}{\Sigma_{yy}}$

Scalar measurement

$y, z \in \mathbb{R} \quad z \sim \mathcal{N}(\mu_z, \sigma_z^2)$
$y \mid z \sim \mathcal{N}(z, \sigma_\varepsilon^2)$
σ_ε^2 is measurement-noise variance.

$$\mu_y = \mu_z \quad \Sigma_{zy} = \sigma_z^2 \quad \Sigma_{yy} = \sigma_z^2 + \sigma_\varepsilon^2$$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2}$$

$$\mu_{z|y} = \mu_z + K(y - \mu_z)$$

Precise measurement

$\sigma_\varepsilon^2 \ll \sigma_z^2 \quad K \approx 1$



Trust the measurement

Equal uncertainty

$\sigma_\varepsilon^2 = \sigma_z^2 \quad K = \frac{1}{2}$



Split the difference

Noisy measurement

$\sigma_\varepsilon^2 \gg \sigma_z^2 \quad K \approx 0$



Trust the prior

- K is the fraction of the prediction error used to update the estimate
- The orange diamond marks the posterior mean $\mu_{z|y}$
- More measurement noise gives smaller K; more prior uncertainty gives larger K

Inference often combines conditioning and summarizing

Not built yet

Sensor fusion combines measurements according to their uncertainty

Not built yet

The same model can also predict measurements

Not built yet

Summary: one model, three probability operations

Not built yet